

Absolute Asymmetric Synthesis of Enantiopure Organozinc Reagents, Followed by Highly Enantioselective Chlorination

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Abstract: We report the absolute asymmetric synthesis (AAS) of indenylzinc reagents by using total spontaneous resolution followed by enantiospecific conversion into 1-chloroindene. The chiral complex [Zn(dcp)(ind)(tmeda)] (dcp = 2,6-dichlorophenoxy and tmeda = *N,N,N',N'*-tetramethylethylenediamine) (**3**) was prepared from the achiral starting materials indene, potassium, zinc chloride, TMEDA, and 2,6-dichlorophenol. The reagent resolved spontaneously on crystallization, and

single crystals of **3** react with *N*-chlorosuccinimide in the presence of benzoquinone in 2-propanol to give 1-chloroindene in >98% enantiomeric excess. It was found that (*R*)-**3** gave (*R*)-1-chloroindene upon reaction, indicating an S_E2'-mechanism. Since bulk samples

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of **3** gave optically active product upon chlorination, total spontaneous resolution must have occurred. This demonstrates that enantiopure products can be obtained through the absolute asymmetric synthesis of organometallic reagents starting from achiral materials. The general absolute asymmetric synthesis (AAS) method offers easy access to both enantiomers and an almost limitless variation in the design of the product.

Introduction

The synthesis of optically active compounds from achiral or racemic precursors, without the influence of pre-formed optical activity, for example, chiral catalysts or auxiliaries, is known as absolute asymmetric synthesis.^[1–4] There are only a few reported genuine examples of absolute asymmetric synthesis in the literature. Optical activity is crucial to living systems, and new examples of absolute asymmetric synthesis may be relevant for the origin of biomolecular homochirality.^[5,6] In addition, absolute asymmetric synthesis has a potential to become a new useful method in inorganic and organic synthesis. One approach to absolute asymmetric synthesis is total spontaneous resolution, which is a method based on selective crystallization of one enantiomer coupled with a rapid interconversion of enantiomers in solution (or melt).^[7–10] Spontaneous resolution occurs when a racemate crystallizes in one of the Sohncke space groups^[11] and results in a mechanical mixture of enantiomerically pure crystals, known as a conglomerate. The maximum yield of a single enantiomer is restricted to 50%. If spontaneous resolution is combined with rapid enantiomerization in solution, a quantitative yield of one single enantiomer may be ob-

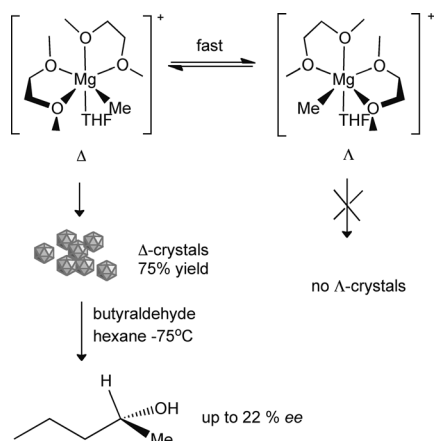
tained; this requires that all crystals grow from a single nucleus through secondary nucleation. A similar evolution from achirality to a single handedness in bulk samples can also take place under mechanical grinding of racemic mixtures of conglomerate crystals in contact with their saturated solutions, as was first reported by Viedma in 2005.^[12–14] The compounds in these examples may either be achiral in solution or made to racemize catalytically or through a chemical reaction. The creation of optical activity during a crystallization is referred to as total spontaneous resolution or crystallization-induced asymmetric transformation^[15] and may be regarded as a form of absolute asymmetric synthesis. We have recently reported total spontaneous resolution of a number of different compounds, for example, organometallic reagents,^[16–19] an aluminium phenoxide complex,^[20] a coordination network,^[21] a conformationally chiral sulfide,^[22] an octanuclear organozinc complex,^[23] a helical coordination polymer,^[24] and several coordination compounds for example, five-,^[25] seven-^[26] and nine-coordinate complexes.^[27] The stereochemical lability, which is required for total spontaneous resolution to occur, means that the optically active product will racemize on dissolution. The optical activity may, however, be fixed through a reaction that yields a stereochemically inert product. Such a reaction has to be performed under conditions that prevent enantiomerization (e.g., solvent-free reactions). The majority of the reported examples of such transformations use photochemical intracrystal reactions.^[28–31] Although many of these reactions give very high stereoselectivity, they suffer from being limited to specific substrates or reagents. The first reaction between an optically active crystal obtained through spontaneous resolution and a chemical reagent was reported in 1969: crystalline

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4,4'-dimethylchalcone gave the dibromide with 6% enantiomeric excess (*ee*) upon reaction with bromine vapor.^[7]

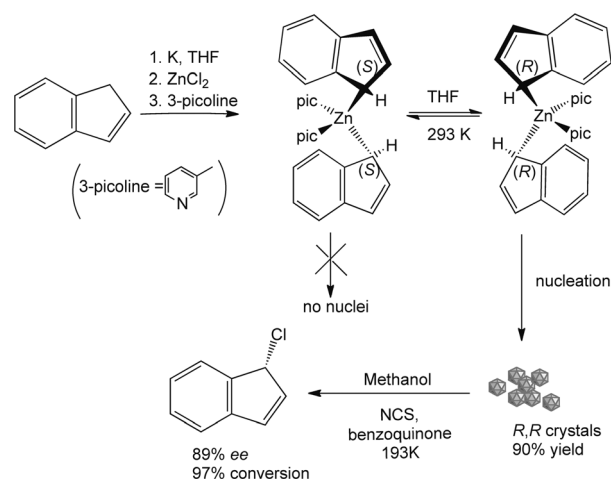
Our approach to the problem of limited product scope is to perform absolute asymmetric synthesis of organometallic reagents, since these are useful in the synthesis of a wide variety of stereochemically inert organic compounds. As a first step in combining absolute asymmetric synthesis with organometallic chemistry, a series of octahedral chiral-at-metal Grignard reagents were prepared, two of which crystallized as conglomerates.^[17] For example, crystal batches of *cis*-[MgCH₃(thf)(dme)₂]I (thf = tetrahydrofuran and dme = 1,2-dimethoxyethane) gave optically active alcohols with 0–22% *ee* upon reaction with prochiral aldehydes (Scheme 1).



Scheme 1. Reaction between a “chiral-at-magnesium” Grignard reagent and an aldehyde to provide a chiral alcohol with 0–22% *ee*.

Both enantiomers of the alcohols could be obtained depending on which enantiomer of the Grignard reagent was used. This was the first report of total spontaneous resolution of an organometallic reagent with concomitant transfer of the optical activity. The low enantiomeric excess is possibly explained by the need to form a new chirogenic centre during the reaction. We therefore directed our efforts at synthesizing organometallic reagents with chirogenic α -carbon atoms, since this is a rare case in which a chirogenic carbon center may become stereochemically labile. When such a reagent reacts with an electrophile, the chirogenic center in the organometallic reagent may remain intact, thus eliminating the need to transform helical chirality to central chirality. Our first attempts involved substituted benzyl lithium complexes^[19] and *sec*-butylzinc complexes.^[32] To overcome difficulties encountered in handling extremely air-sensitive solid organolithium reagents, and the low rate of enantiomerization of dialkylzinc reagents, we turned to diindenylzinc complexes. These reagents may enantiomerize through an η^3 intermediate or a rapid [1,3]-Zn shift in solution, whereas in the solid state zinc is prone to be σ -bonded to the indenyl ligand in a monohapto fashion giving rise to a chirogenic α -carbon atom. In search for a conglomerate, a number of diindenylzinc complexes were prepared, and it was found that [Zn(ind)₂(pic)₂] (ind = η^1 -indenyl and pic = 3-

picoline) underwent spontaneous resolution.^[18] When enantiomerically pure single-crystals of [Zn(ind)₂(pic)₂] were ground with *N*-chlorosuccinimide (NCS) cooled in liquid nitrogen followed by quenching with water, 1-chloroindene with a modest *ee* of 17–22% was obtained. Adding methanol and NCS to a crystal of [Zn(ind)₂(pic)₂] in a phial immersed in liquid nitrogen, removing the phial from the cooling bath and grinding until the mixture had reached ambient temperature increased the yield of 1-chloroindene and the enantiomeric excess increased to 26–29% *ee*. Other solvents, such as THF and acetonitrile, gave considerably lower enantiomeric excess. Finally, addition of the radical inhibitor 1,4-benzoquinone to the NCS prior to reaction gave a dramatic increase in the enantiomeric excess giving 82–89% *ee* and 78–99% yield (Scheme 2). Whole crystal batches obtained



Scheme 2. Total spontaneous resolution of an organozinc reagent in one solvent, followed by chirality transfer to afford an enantioenriched product in a different solvent.

through total spontaneous resolution gave up to 71% *ee*. In addition, by using racemic [Zn(ind)₂(pic)₂], this reaction is more convenient than literature methods for the synthesis of racemic 1-chloroindene.^[33] Both enantiomers of 1-chloroindene can be obtained by the absolute asymmetric synthesis method and may be useful in organic synthesis since organic halides are versatile building blocks for constructing larger molecules.^[34–36] Stereochemical control is possible by seeding with the desired enantiomorph.

The presence of two indenyl groups in [Zn(ind)₂(pic)₂] may have reduced the overall *ee* value upon reaction with NCS since an intermediate species may rearrange to adopt a higher hapticity (with loss of chirality) before reacting with a second equivalent of NCS. In addition, the presence of a concomitantly crystallizing racemic phase *rac*-[Zn(ind)₂(pic)₂] and the haphazardous formation of crystals that are twinned by inversion were additional disturbing factors that urged us to modify the reagent. Replacing one of the indenyl groups with an inert anionic ligand could give a reagent with improved selectivity if a conglomerate phase is found. We now report the results regarding the synthesis, crystalli-

zation, and chlorination of organozinc reagents bearing a single indenyl ligand.

Results and Discussion

Synthesis and spontaneous resolution: Complexes of the type $[\text{Zn}(\text{X})(\text{ind})(\text{L})_n]$, in which X is an anionic ligand and L is a neutral ligand, can be obtained by addition of one equivalent of an acid, HX, to diindenylzinc in THF, followed by addition of the neutral ligand L. With two variables (X and L), synthesis of a large number of complexes is possible, and there should be a good chance to find a complex crystallizing as a conglomerate. Initial experiments with indenylzinc halide complexes invariably resulted in Schlenk-type equilibria and crystallization of zinc dihalide complexes; consequently we turned to indenylzinc phenoxide complexes. For example, 2,6-dichlorophenol readily protonates and displaces one indenyl ligand from diindenylzinc resulting in a colour change from orange to pale yellow. Subsequent evaporation of the solvent gave colorless crystals of the dinuclear complex *meso*- $[\{\text{Zn}(\text{dcp})(\text{ind})(\text{thf})\}_2]$ (dcp = 2,6-dichlorophenoxy and thf = tetrahydrofuran), **1**, (Figure 1).

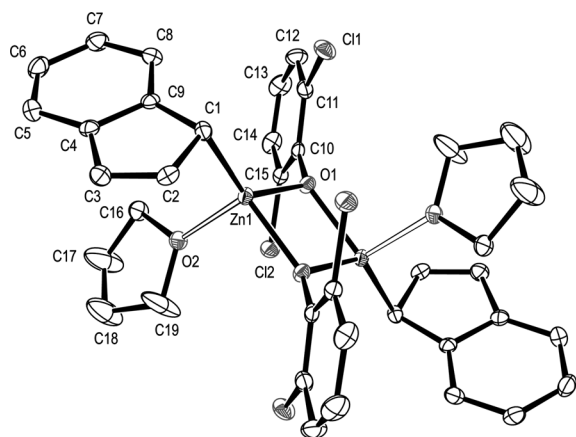


Figure 1. Molecular structure of *meso*- $[\{\text{Zn}(\text{dcp})(\text{ind})(\text{thf})\}_2]$ (**1**) showing the crystallographic numbering scheme. Displacement ellipsoids are drawn at the 50% probability level. All H atoms have been omitted for clarity. Selected bond lengths [Å] and angles [°]: Zn1–C1 2.012(3), Zn1–O1 2.010(2), Zn1–O2 2.073(2), C1–Zn1–O1 130.98(9), C1–Zn1–O2 109.98(11), O1–Zn1–O2 97.64(8).

The 2,6-dichlorophenoxide ligands in **1** bridge between the two zinc centers, which display distorted tetrahedral coordination geometries. The indenyl groups are coordinated in a η^1 fashion and thus display chirogenic α -carbon atoms, but the two indenyl ligands have opposite configurations; the complex is thus a *meso* form and would fail to give an *ee* value upon reaction with an electrophile even if a conglomerate phase was found. Complex **1** crystallizes in the centrosymmetric space group $P\bar{1}$ with the molecules located at a crystallographic inversion center. To obtain mononuclear complexes (thus avoiding the *meso* problem), solutions of **1**

were treated with bidentate ligands. In this way crystals of $[\text{Zn}(\text{dcp})(\text{ind})(\text{teeda})]$ (teeda = *N,N,N',N'*-tetraethylethylenediamine), **2** (Figure 2), were obtained after crystallization from toluene.

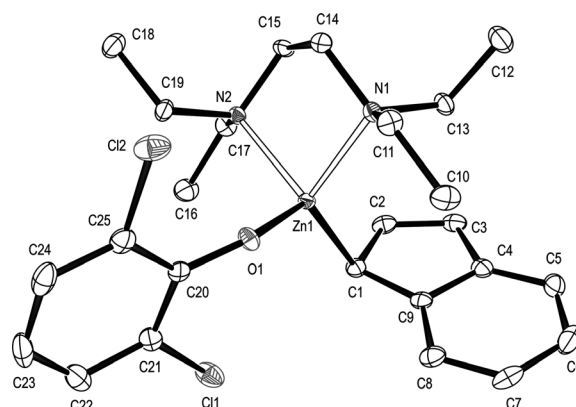


Figure 2. Molecular structure of *rac*- $[\text{Zn}(\text{dcp})(\text{ind})(\text{teeda})]$ (**2**) showing the crystallographic numbering scheme. Displacement ellipsoids are drawn at the 50% probability level. All H atoms have been omitted for clarity. Selected bond lengths [Å] and angles [°]: Zn1–O1 1.9526(12), Zn1–N1 2.1364(14), Zn1–N2 2.1463(14), Zn1–C1 2.0706(17), C1–Zn1–N1 119.97(6), C1–Zn1–N2 118.09(6), C1–Zn1–O1 123.95(6), O1–Zn1–N1 97.24(5), O1–Zn1–N2 102.89(5), N1–Zn1–N2 87.18(5).

The 2,6-dichlorophenoxide ligand in **2** is not bridging, and a mononuclear, chiral complex results. Despite being chiral, **2** crystallizes in the centrosymmetric space group $P2_1/n$ and thus forms racemic crystals. Small modifications of the molecular structure usually cause large alterations in the crystal structure, and we were pleased to find that replacing TEEDA with TMEDA (*N,N,N',N'*-tetramethylethylenediamine) gave the sought conglomerate: $[\text{Zn}(\text{dcp})(\text{ind})(\text{tmeda})]$, (**3**) crystallized in the Sohncke space group $P2_12_12_1$ (Figure 3). Under the crystallization conditions employed no racemic phase of $[\text{Zn}(\text{dcp})(\text{ind})(\text{tmeda})]$ was ever observed, which is noteworthy since it is not uncommon to obtain both chiral and racemic phases, and the two phases may even crystallize concomitantly.^[18,22,37] Replacing DCP with 2,6-dibromophenol (DBP) gave crystals of $[\text{Zn}(\text{dbp})(\text{ind})(\text{tmeda})]$, **4**, (Figure 4), which is isomorphous with **3** and thus also crystallizes as a conglomerate. The absolute structures of **3** and **4** are unambiguously determined, as indicated by the low Flack parameter,^[38,39] $-0.007(7)$ for **3** and $-0.010(12)$ for **4**. No problems with twinning by inversion were encountered.

As for **1**, the coordination of the indenyl group in **2**, **3** and **4** is close to η^1 . This is manifested in significantly longer Zn1–C2 and Zn1–C9 distances than in Zn–C1. Also, as seen from Table 1, there is a significant localization of double bonds in the five-membered ring, indicating a high degree of Zn–C σ -bonding: the C2–C3 distances are generally shorter than the C1–C2 and C3–C4 distances. Compared with **2**, the Zn1–C1 distances in **3** and **4** are slightly shorter, perhaps due to steric hindrance by the larger TEEDA ligand in **2**. The chelate rings in **2**, **3**, and **4** adopt

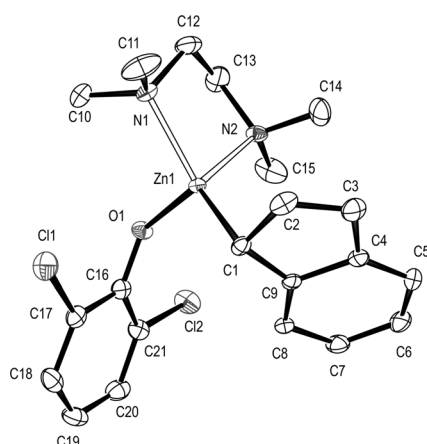


Figure 3. Molecular structures of *(R)*-[Zn(dcp)(ind)(tmeda)] (*R*-**3**) (left) showing the crystallographic numbering scheme. Displacement ellipsoids are drawn at the 50% probability level. All H atoms have been omitted for clarity. Selected bond lengths [Å] and angles [°]: Zn1–O1 1.9419(12), Zn1–N1 2.1389(16), Zn1–N2 2.1096(16), Zn1–C1 2.0465(17) C1–Zn1–N1 125.40(7), C1–Zn1–N2 119.29(7), C1–Zn1–O1 115.04(7), O1–Zn1–N1 104.01(6), O1–Zn1–N2 101.80(6), N1–Zn1–N2 85.89(7)

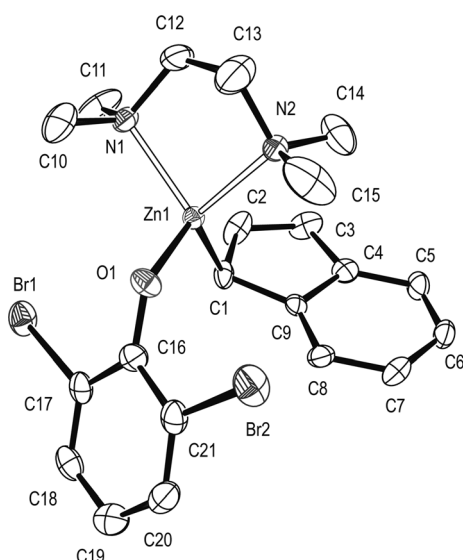
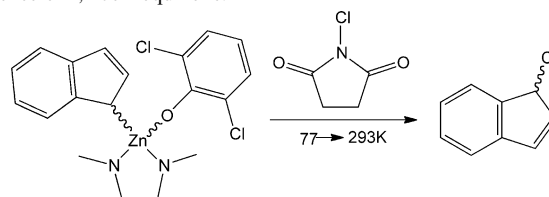


Figure 4. Molecular structures of *(S)*-[Zn(dbp)(ind)(tmeda)] (*S*-**4**) showing the crystallographic numbering scheme. Displacement ellipsoids are drawn at the 50% probability level. All H atoms have been omitted for clarity. Selected bond lengths [Å] and angles [°]: Zn1–O1 1.952(3), Zn1–N1 2.133(3), Zn1–N2 2.113(3), Zn1–C1 2.041(4) C1–Zn1–N1 123.49(15), C1–Zn1–N2 118.96(14), C1–Zn1–O1 113.11(14), O1–Zn1–N1 105.38(12), O1–Zn1–N2 105.94(12), N1–Zn1–N2 86.09(13)

chiral conformations (thus adding an additional element of chirality to these complexes): exclusively δ for (*S*)-**2**, (*S*)-**3**, or (*S*)-**4** and exclusively λ for (*R*)-**2**, (*R*)-**3**, and (*R*)-**4**. The shortest intermolecular interaction in **3** (Figure 5) is a CH–O interaction of 2.452(1) Å between H3 and O1ⁱ (symmetry code (i): $3/2-x, 2-y, -1/2+z$). These interactions give rise to chains extending parallel to the *c* axis. An analogous set of interactions are found in the isomorphous **4**. By contrast, no strong intermolecular interactions are observed in **1** or **2**.

Table 1. Formation of 1-chloroindene from crystals of **3** and NCS in the presence of 1,4-benzoquinone.



Entry	Reagent	Solvent	Product rotation	Conversion ^[a] [%]	ee [%]
1	(<i>S</i>)	methanol	(+)	71–94	38–96
2	(<i>S</i>)	ethanol	(+)	65–97	78–98
3	(<i>S</i>)	acetonitrile	(+)	79–99	3–19
4	(<i>S</i>)	water	(+)	73–86	52–76
5	(<i>S</i>)	THF	– ^[b]	N.D. ^[c]	0
6	(<i>R</i>)	acetone	(–)	N.D. ^[c]	44–55
7	(<i>R</i>)	1-propanol	(–)	N.D. ^[c]	72–93
8	(<i>S</i>)	ethylene glycol	(+)	75–99	76–85
9	(<i>S</i>)	2-propanol	(+)	92–97	92–98
10	(<i>S</i>)	2-propanol ^[d]	(+)	95–98	50–51
11	(<i>R</i>)	2-propanol ^[e]	(–)	20–89	0–94
12	(<i>S</i>)	2-propanol ^[e]	(+)	20–95	0–95
13 ^[f]	(<i>R</i>)	2-propanol	(–)	89	59

[a] Determined by HPLC on a Chiralcel OD stationary phase and confirmed by integration of the ALP chromatogram. [b] No 1-chloroindene formed. [c] Not determined due to formation of unidentified byproducts. [d] No benzoquinone added. [e] Dominant enantiomer, full batch. [f] Addition at ambient temperature.

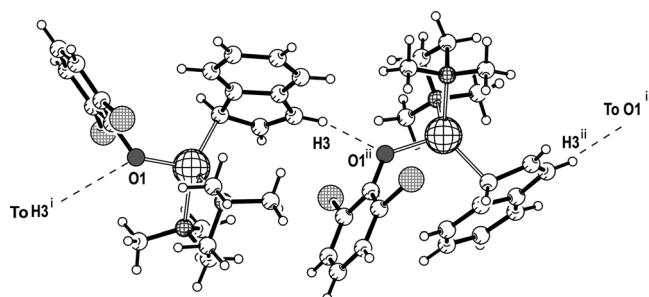
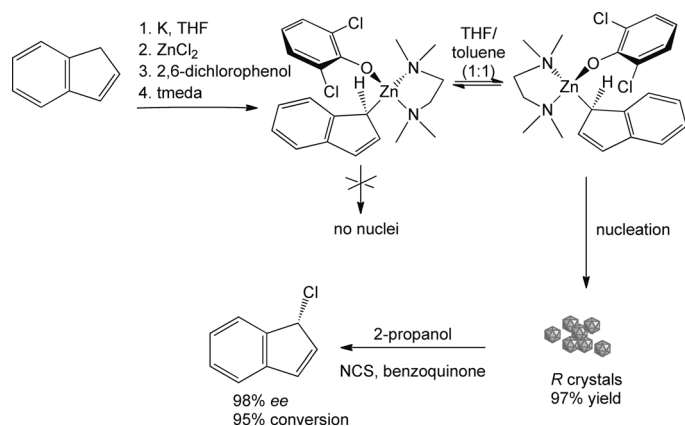


Figure 5. The shortest intermolecular interaction in the crystal structure of **3** is the H3–O1 interaction. This results in parallel chains of molecules running along the crystallographic *c* axis (symmetry code (i): $3/2-x, 2-y, 1/2+z$, (ii) $3/2-x, 2-y, -1/2+z$).

The fact that compounds **3** and **4** exhibit monohapto coordination with a single chirogenic α -carbon atom in each molecule and, in addition, crystallize in a Sohncke space group with low Flack parameters, makes them excellent candidates for absolute asymmetric synthesis.

Treatment of single crystals of 3 with NCS: To evaluate the enantiomeric excess in full crystal batches we first needed to find a reaction that would reproducibly give a stereochemically inert product with high *ee* value from single crystals of **3** or **4**. Large single crystals of **3** were obtained more easily than crystals of **4** and compound **3** was therefore used in studies of enantioselective reactions. The test reactions were performed on single crystals, since these were known from X-ray diffraction experiments to be enantiomerically pure.

When single crystals of **3** were subjected to the same reaction conditions that was found ideal for crystals of $[\text{Zn}(\text{ind})_2(\text{pic})_2]$,^[18] (grinding the crystal at -196°C , slow addition of methanol followed by NCS and benzoquinone and grinding while allowing the mixture to reach ambient temperature) it was evident that these conditions needed modification. One single crystal, confirmed by X-ray diffraction to be enantiomerically pure, gave 1-chloroindene with a mediocre 45% *ee*. An increase of up to 96% *ee* was observed when the NCS/benzoquinone mixture was added to a crushed crystal as a suspension in methanol, but the highest selectivity could not be consistently reproduced. This finding prompted further refinement of the reaction conditions, since chlorination of **3** seemed more dependent on the mode of addition, compared with crystals of $[\text{Zn}(\text{ind})_2(\text{pic})_2]$. Surprisingly, replacing methanol by ethanol increased the maximum *ee* to 98% and using 2-propanol gave even better results; at best >98% *ee* with 95% conversion (Scheme 3).



Scheme 3. Total spontaneous resolution of **3**, followed by highly stereoselective chlorination with NCS.

As for $[\text{Zn}(\text{ind})_2(\text{pic})_2]$, the *ee* value obtained for 1-chloroindene after reaction with NCS was highly dependent on the solvent (Table 1). Chlorination of **3** in water gave widely varying outcomes, whereas the use of ethylene glycol gave conversions and *ee* values similar to those obtained with methanol. Acetone and 1-propanol gave unidentified by-products. Acetonitrile never gave more than 19% *ee* even though the conversion was excellent, whereas the use of THF completely failed to give 1-chloroindene. Based on these findings, 2-propanol was chosen for further optimization of the reaction. Adding a suspension of NCS and benzoquinone in 2-propanol to single crystals of **3** gave somewhat erratic results, but when a dilute solution was employed the outcome became reproducible (Table 2). Thus adding 2-propanol (0.2 mL) containing (NCS 10 mg) and benzoquinone (3 mg) to each of 10 crystals of **3** gave 1-chloroindene in 91–98% *ee* and 92–97% conversion (entry 5 in Table 2 most likely featured two aggregated crystals of opposite chirality). Just as for $[\text{Zn}(\text{ind})_2(\text{pic})_2]$, the strong

Table 2. Results of chlorination of **3** using a solution of NCS and 1,4-benzoquinone in 2-propanol.

Entry	Reagent	Product rotation	Conversion [%]	<i>ee</i> [%]
1	(S)	(+)	95	98
2	(R)	(−)	96	93
3	(S)	(+)	95	96
4	(S)	(+)	92	98
5	(R)	(−)	96	51
6	(R)	(−)	97	92
7	(S)	(+)	92	97
8	(S)	(+)	95	98
9	(R)	(−)	94	96
10	(S)	(+)	95	96
11	(S)	(+)	95	94
12	(R)	(−)	96	94
13	(R)	(−)	93	93
14	(R)	(−)	95	95

dependence on the solvent indicate a reaction in solution. Grinding the solid reagents with crystals of **3** at low temperature, followed by quenching with gaseous HCl, resulted in <15% 1-chloroindene, which further points to a reaction in solution. The chlorination must occur rapidly after dissolution, as it was shown by ^1H NMR spectroscopy in C_6D_6 at ambient temperature that the protons on C1 and C3 are equivalent in both $[\text{Zn}(\text{ind})_2(\text{pic})_2]$ and **3**. This means that the zinc atom in both complexes have either η^3 or η^5 hapticity or is in rapid exchange and thus should give racemic 1-chloroindene upon reaction with NCS in solution at ambient temperature. As expected, performing the reaction with NCS at ambient temperature led to a decrease in both the *ee* value and conversion, however this was not as significant as could have been expected (Table 1, entry 13). The reaction most likely takes place through an $\text{S}_{\text{E}}2'$ -mechanism with attack on C3 and double-bond shift. Such a mechanism implies that (*R*)-**3** would give (*R*)-1-chloroindene. To investigate this, the absolute structure of a crystal of **3** was determined by X-ray diffraction and the crystal was then treated with NCS. A CD-spectrum was recorded for the 1-chloroindene obtained from a single crystal of known absolute structure, that is, (*R*)-**3**, and this spectrum was compared to the calculated spectrum for (*R*)-1-chloroindene. The matching signs for the Cotton effects indicated that the reaction indeed occurs with retention of the configuration at C1.

It was also found that crystals of the related $[\text{Zn}(\text{meind})_2(\text{tmeda})]$ (meind=3-methylindenyl) (**5**) (Figure 6) gave 1-chloro-1-methylindene as the single product on reaction with NCS in methanol. Keeping in mind that the steric and electronic effects arising from a substituent in 3-position may affect the mechanism significantly, it was clear that at least this related reagent reacted exclusively through an $\text{S}_{\text{E}}2'$ pathway. Since **5** crystallizes in the centrosymmetric space group $P2_1/n$, 1-chloro-1-methylindene was obtained as a racemate.

^1H NMR spectroscopy revealed that the reaction mixture obtained after treatment of **3** and NCS, contained 1-chloroindene, traces of indene, and the succinimidato complex

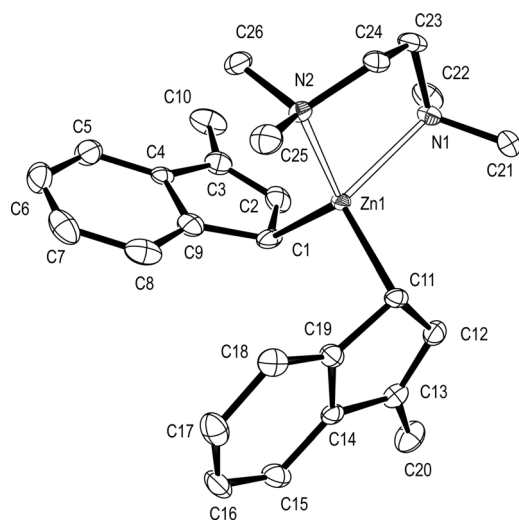


Figure 6. Molecular structure of (*R*)-**5** showing the crystallographic numbering scheme. Displacement ellipsoids are drawn at the 50% probability level. All H atoms have been omitted for clarity. Selected bond lengths [Å] and angles [°]: Zn1–C1 2.120(2), Zn1–N1 2.205(2), Zn1–N2 2.132(2), Zn1–C11 2.069(3), C1–Zn1–C11 112.48(10), N1–Zn1–N2 84.11(8), C1–Zn1–N1 109.12(9), C1–Zn1–N2 109.15(9), C11–Zn1–N2 119.93(10)

[Zn(dcp)(succinimidato)(tmeda)], **6**, (Figure 7). Reference crystals of **6** were obtained by mixing diethylzinc, succinimide, and TMEDA. The crystal structure of **6** was determined (Figure 7) and the signals from the ^1H NMR spectrum of dissolved crystals of **6** were identical to those found in the spectrum obtained from the reaction of **3** with one equivalent of NCS. When bulk amounts of **3** were used rather than single crystals, an unidentified red insoluble tar-like byproduct was also obtained, but was easily separated during work-up. The reaction of $[\text{Zn}(\text{ind})_2(\text{pic})_2]$ is always

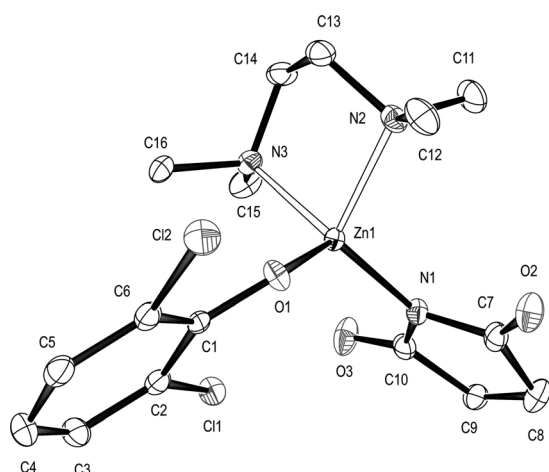


Figure 7. Molecular structure of **6** showing the crystallographic numbering scheme. Displacement ellipsoids are drawn at the 50% probability level. All H atoms have been omitted for clarity. Selected bond lengths [Å] and angles [°]: Zn1–O1 1.9176(16), Zn1–N1 1.9654(19), Zn1–N3 2.094(2), Zn1–N2 2.122(2), O1–Zn1–N1 121.73(8), O1–Zn1–N3 107.95(7), N1–Zn1–N3 122.80(8), O1–Zn1–N2 95.89(7), N1–Zn1–N2 112.14(8), N3–Zn1–N2 87.21(8)

free from unidentified byproducts. We also re-examined the chlorination of $[\text{Zn}(\text{ind})_2(\text{pic})_2]$ with two equivalents of NCS and analyzed the ^1H NMR spectrum from this reaction mixture to detect $[\text{Zn}(\text{succinimidato})_2(\text{pic})_2]$ (**7**). Reference crystals were synthesized from diethylzinc, succinimide and 3-methylpyridine in THF. Crystallization from THF gave $[\text{Zn}(\text{succinimidato})_2(\text{pic})_2]\cdot\text{THF}$ (**8**) featuring co-crystallized THF molecules (Figure 8), whereas crystallization from di-

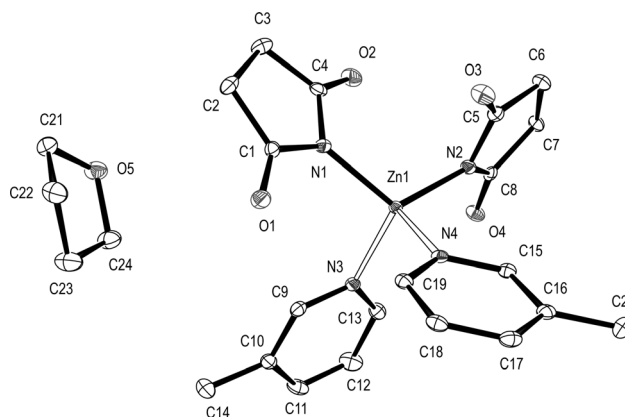


Figure 8. Molecular structure of **8** showing the crystallographic numbering scheme. Displacement ellipsoids are drawn at the 50% probability level. All H atoms have been omitted for clarity. Selected bond lengths [Å] and angles [°]: Zn1–N1 1.9796(17), Zn1–N2 1.9703(16), Zn1–N3 2.0679(17), Zn1–N4 2.0690(17), N2–Zn1–N1 114.38(7), N2–Zn1–N3 118.35(7), N1–Zn1–N3 105.22(7), N2–Zn1–N4 103.73(7), N1–Zn1–N4 115.39(7), N3–Zn1–N4 99.19(6)

ethyl ether rendered the solvent-free complex **7**. The crystals of **7** were of poor quality but the composition of the complex could be determined from the crystal structure. The signals in the ^1H NMR spectrum of **7** could indeed be found along with signals from 1-chloroindene in the spectrum from the reaction mixture of $[\text{Zn}(\text{ind})_2(\text{pic})_2]$ with two equivalents of NCS.

Absolute asymmetric synthesis: The *ee* value of full crystal batches can be determined by solid state CD spectroscopy, but only for air-stable compounds.^[26] For air-sensitive compounds such as **3** and **4**, establishing whether the whole crystal batch is optically active is more difficult. However, after having optimized the reaction of single crystals with NCS, we had a convenient method for determining the *ee* value of full crystal batches. Absolute asymmetric synthesis is achieved through total spontaneous resolution of **3**. On one occasion, leaving a supersaturated solution of **3** at ambient temperature resulted in one large single crystal with 55% yield, which gave 1-chloroindene with 95% *ee*. This was not, however, a reproducible method, since at the concentration necessary for crystals to form, the risk of several simultaneous primary nucleation events was very high. Instead, slow evaporation of the solvents from a solution of **3** in THF/toluene (1:1) under rapid stirring^[40,41] was investigated. This gave a light-yellow precipitate in quantitative yield. Upon reaction with NCS and 1,4-benzoquinone in 2-propanol the

batches gave 1-chloroindene with *ee* values ranging between 0 and 95 %. Many batches were crystallized and no bias for either *R*- or *S*-batches could be found. An *ee* value of 95 % means that the whole batch must have been close to enantiomerically pure, and to our knowledge this is the highest *ee* value demonstrated for a reaction using an entire batch of a chiral organometallic reagent obtained through total spontaneous resolution (TSR). A promising alternative to TSR for obtaining enantiopure crystal batches would be attrition-enhanced (Viedma) deracemization.^[12–14, 42] However, preliminary grinding experiments of indenylzinc reagents have been unsuccessful in our hands.

Conclusion

We have demonstrated how enantiopure 1-chloroindene (a stereochemically inert product) can be obtained through AAS of indenylzinc reagents, obtained from achiral starting materials. Enantiopure indenylzinc crystal batches could be obtained in large scale by TSR. We propose a general AAS/TSR method that offers easy access to enantiopure organometallic reagents. A requirement for this method is that new conglomerate phases of organometallic reagents can be identified. This should be possible since the nucleophilic group in the reagent can be varied, as well as the metal and the neutral ligand. A new conglomerate reagent may be able to react with a range of different electrophiles; fine-tuning reactions to give many new products for each reagent should be possible. Both enantiomers will be equally accessible and the potential of the AAS method in organic synthesis should be substantial.

Experimental Section

General considerations: All experiments were performed by using standard Schlenk equipment under a nitrogen atmosphere. All glassware was dried in 130 °C overnight. Commercial diethylzinc (15 % in hexane, Acros) and succinimide (Eastman Organic Chemicals) was used as received. Tetrahydrofuran (THF), diethyl ether (DEE) and toluene were distilled from sodium/benzophenone prior to use. Indene (Aldrich) was distilled prior to use. Zinc chloride (Alfa Aesar) was dried using thionyl chloride.^[43] Tetramethylethylenediamine (Aldrich), tetraethylethylenediamine (Aldrich) and 3-methylpyridine (Aldrich) were distilled from CaH₂ under reduced pressure and stored under a nitrogen atmosphere. 1,4-Benzoquinone (Aldrich) was sublimed under reduced pressure at ambient temperature. The determination of the *ee* values were performed using a Varian model 9000 HPLC system combined with a PDR advanced laser polarimeter. A Daicel OD column was used. ¹H NMR spectra were recorded on a JEOL Eclipse 400 instrument operating at 400 MHz.

X-ray diffraction: Crystals were selected and mounted under nitrogen in glass capillaries at low temperature^[44] and transferred to a Rigaku R-Axis IIC image plate system. Diffracted intensities were measured at 100(2) K, using graphite-monochromated MoK α radiation from a RU-H3R rotating anode. The structures were solved by using the program SIR-92^[45] and refined using full-matrix least squares calculations on *F*² using the SHELXL-97 program.^[46] All non-hydrogen atoms were refined anisotropically, and hydrogen atoms were included in calculated positions and refined using a riding model. Structures were drawn using ORTEP3^[47] for Windows or PLATON.^[48] SIR-92, SHELXL-97, and ORTEP3 were enclosed in the WinGX-software package.^[49] See Table 3 for the X-ray data for compounds **1–6** and **8**. CCDC-924605(**1**), CCDC-924606(**2**), CCDC-924607((*R*)-**3**), CCDC-924608 ((*S*)-**3**), CCDC-924609 ((*R*)-**4**), CCDC-924610 ((*S*)-**4**), CCDC-924611 (**5**), CCDC-924612 (**6**), and CCDC-924613 (**8**) contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Preparation of meso-[Zn(2,6-dcp)(ind)(thf)]₂ (1**):** A solution of diindenylzinc^[50] (1.58 g, 3.6 mmol) in THF (7.6 mL) was added under stirring to 2,6-dichlorophenol (0.59 g, 3.6 mmol) in THF (2 mL). After 30 min, 1 mL

Table 3. X-ray data for compounds **1**, **2**, (*R*)-**3**, (*R*)-**4**, **5**, **6**, and **8**.

	1	2	(<i>R</i>)- 3	(<i>R</i>)- 4	5	6	8
formula	C ₃₈ H ₃₆ Cl ₄ O ₄ Zn ₂	C ₂₅ H ₂₆ Cl ₂ N ₂ OZn	C ₂₁ H ₂₆ Cl ₂ N ₂ OZn	C ₂₁ H ₂₆ Br ₂ N ₂ OZn	C ₂₅ H ₂₅ N ₂ Zn	C ₁₆ H ₂₅ Cl ₂ N ₃ O ₃ Zn	C ₂₄ H ₃₀ N ₄ O ₅ Zn
<i>M_w</i>	829.32	506.75	458.71	547.63	418.84	441.64	519.89
space group	<i>P</i> $\bar{1}$	<i>P</i> ₂ / <i>n</i>	<i>P</i> ₂ /2 ₁	<i>P</i> ₂ /2 ₁	<i>P</i> ₂ / <i>n</i>	<i>P</i> ₂ / <i>a</i>	<i>P</i> $\bar{1}$
<i>a</i> [Å]	9.1746(13)	10.1945(17)	7.9920(10)	7.9302(12)	10.6718(18)	15.901(3)	8.5272(12)
<i>b</i> [Å]	9.6157(13)	15.573(2)	16.055(2)	16.420(2)	19.163(3)	7.0102(12)	11.8554(10)
<i>c</i> [Å]	11.6806(4)	16.442(3)	16.529(2)	16.744(3)	11.1724(18)	17.957(3)	12.3556(15)
α [°]	62.798(13)	90.00	90	90	90	90	74.649(16)
β [°]	75.88(3)	107.413(6)	90	90	94.121(6)	107.447(6)	86.618(19)
γ [°]	88.93(3)	90.00	90	90	90	90	79.907(13)
<i>V</i> [Å ³]	883.63(18)	2490.7(7)	2120.9(5)	2180.3(6)	2278.9(6)	1909.5(6)	1185.8(2)
<i>Z</i>	1	4	4	4	4	4	2
ρ_{calcd} [g cm ^{−3}]	1.558	1.351	1.437	1.668	1.282	1.536	1.456
μ (MoK α) [mm ^{−1}]	1.700	1.219	1.423	4.805	1.092	1.586	1.079
<i>T</i> [K]	100(2)	100(2)	100(2)	100(2)	100(2)	100(2)	100(2)
crystal size [mm]	0.20 × 0.20 × 0.10	0.20 × 0.15 × 0.10	0.40 × 0.40 × 0.30	0.20 × 0.20 × 0.10	0.30 × 0.20 × 0.15	0.40 × 0.20 × 0.15	0.30 × 0.40 × 0.40
reflns collected	5489	16402	13910	13687	13945	11751	7762
unique reflns (<i>R</i> _{int})	2787 (0.0404)	4604 (0.0269)	3825 (0.0214)	3773 (0.0278)	3866 (0.0287)	3299 (0.0340)	3938 (0.0468)
GOF on <i>F</i> ²	1.050	1.096	1.101	1.108	1.122	1.037	1.054
final <i>R</i> 1(<i>F</i>) ^[a] [<i>I</i> > 2 σ (<i>I</i>)]	0.0361/0.0966	0.0349/0.0921	0.0196/0.0502	0.0292/0.0614	0.0414/0.0859	0.0286/0.0678	0.0286/0.0701
<i>wR</i> 2(<i>F</i> ²) ^[b]							
<i>R</i> 1 ^[a] / <i>wR</i> 2(<i>F</i> ²) ^[b] (all data)	0.0393/0.0986	0.0423/0.0959	0.0208/0.0507	0.0319/0.0624	0.0451/0.0878	0.0403/0.0720	0.0332/0.0714
Flack parameter	–	–	–0.007(7)	–0.010(12)	–	–	–

[a] *R*1(*F*) = $\Sigma(|F_o| - |F_c|) / \Sigma |F_o|$. [b] *wR*2(*F*²) = $[\Sigma(w(F_o^2 - F_c^2)^2) / \Sigma(w(F_o^2)^2)]^{1/2}$.

of this solution was evaporated to dryness and the solid residue was dissolved in a mixture of hot toluene (2 mL) and THF (0.5 mL). Light-yellow crystals were obtained at ambient temperature (0.13 g, 35%). ¹H NMR (400 MHz, C₆D₆): δ = 7.45 (bm, 4H), 7.06 (bm, 4H), 7.01 (s, 2H), 6.99 (s, 2H), 6.96 (bm, 2H), 6.28 (t, 2H), 5.61 (d, 4H), 3.21 (m, 8H), 1.09 ppm (m, 8H).

Preparation of *rac*-[Zn(2,6-dcp)(ind)(teeda)] (2): A solution of diindenylzinc^[50] (1.58 g, 3.6 mmol) in THF (7.6 mL) was added under stirring to 2,6-dichlorophenol (0.59 g, 3.6 mmol) in THF (2 mL). After 10 min TEEDA (0.76 mL, 3.6 mmol) was added and the mixture was stirred for another 10 min. The solvent from a portion of the solution (1 mL) was removed under reduced pressure and replaced by toluene (2 mL) and the solids were dissolved by heating. Large, yellow crystals in 49% yield (0.11 g) were deposited overnight. ¹H NMR (400 MHz, C₆D₆): δ = 7.71–7.74 (m, 2H), 7.33 (d, 2H), 7.27–7.29 (m, 1H), 7.21–7.23 (m, 2H), 6.38 (t, 1H), 5.91 (bs, 2H), 3.10–3.19 (m, 2H), 2.06–2.15 (m, 2H), 1.95–2.03 (m, 2H), 1.68–1.76 (m, 2H), 1.61–1.68 (m, 2H), 0.54 (t, 6H), 0.44 ppm (t, 6H).

Preparation of (S)- or (R)-[Zn(2,6-dcp)(ind)(tmeda)] (3): A solution of diindenylzinc^[50] (1.58 g, 3.6 mmol) in THF (7.6 mL) was added under stirring to 2,6-dichlorophenol (0.59 g, 3.6 mmol) in THF (2 mL). After 10 min TMEDA (0.54 mL, 3.6 mmol) was added and the mixture stirred for another 10 min. The solvent from a portion of the solution (1 mL) was removed under reduced pressure and replaced by toluene (2 mL) and the solids were dissolved by heating. Large, yellow needle-shaped crystals in 60% yield (0.099 g) were deposited overnight. ¹H NMR (400 MHz, C₆D₆): δ = 7.57–7.62 (m, 2H, C^{5,8}_{Ar}H), 7.27 (d, 2H, C^{18,20}_{Ar}H), 7.14–7.17 (m, 2H, C^{6,7}_{Ar}H), 7.07–7.09 (m, 1H, C¹⁹_{Ar}H), 6.24 (t, 1H, C²H), 5.78 (d, 2H, C^{1,3}H), 1.94 (s, 6H, N-CH₃), 1.48–1.54 (m, 2H, N-CH₂), 1.22 (s, 6H, N-CH₃), 1.07–1.12 ppm (m, 2H, N-CH₂).

Full batches of 3 for absolute asymmetric synthesis: Toluene (1 mL) was added to a solution of 3 in THF (1 mL, 0.47 mmol) and the solvents were removed slowly (1–2 h) under rapid stirring by flushing nitrogen through a cannula inserted through a rubber septum and letting the solvents evaporate through a small opening. The process was continued until all solvents had been removed. The resulting pale-yellow precipitate (0.215 g, 97%) was washed with cold toluene and hexane and dried in vacuo.

Treatment of single crystals of 3 with NCS: A hand-picked single-crystal of 3 was transferred to a small phial under a nitrogen atmosphere and carefully ground to a powder. A suspension of NCS (30 mg) and 1,4-benzoquinone (10 mg) in 2-propanol (0.6 mL) was prepared and 0.2 mL of this solution was slowly added to the crystal while the vial was cooled in liquid nitrogen. The mixture was allowed to slowly reach ambient temperature under constant swirling whereby it gradually became less turbid until a clear-yellow solution was obtained. Hydroquinone (10 mg) was added and the solvent was removed under reduced pressure. The solids were extracted with hexane and the hexane was washed with 2.5 M HCl. The *ee* value and conversion were determined by HPLC and ALP and are not isolated yields. 1-chloroindene: ¹H NMR (400 MHz, CDCl₃): δ = 7.51 (d, 1H, ArCH), 7.21–7.28 (m, 3H, ArCH), 6.80 (d, 1H, H3), 6.37 (d, 1H, H2), 5.28 ppm (s, 1H, H1).

Reaction of full batches of 3 with NCS: A suspension of NCS (50 mg) and 1,4-benzoquinone (25 mg) in 2-propanol (10 mL) was slowly added to the precipitate of 3 (0.11 g, 0.19 mmol) while the reaction vessel was cooled in liquid nitrogen. The mixture was allowed to slowly reach ambient temperature under constant swirling whereby it gradually became less turbid and darkened to an orange solution. On standing at ambient temperature the solution darkened even more and a red-brown solution was formed. Hydroquinone (50 mg) was added and the solvent removed under reduced pressure. The solids were extracted with hexane until the extracts were colorless and the combined organic phases were washed with 2.5 M HCl. The *ee* value and conversion were determined by HPLC and ALP and are not isolated yields.

Preparation of (S)- or (R)-[Zn(2,6-dbp)(ind)(tmeda)] (4): A solution of diindenylzinc^[50] (1.58 g, 3.6 mmol) in THF (7.6 mL) was added under stirring to 2,6-dibromophenol (0.34 g, 3.6 mmol) in THF (2 mL). After 10 min TMEDA (0.54 mL, 3.6 mmol) was added and the mixture stirred

for another 10 min. The solvent from a portion of the solution (1 mL) was removed under reduced pressure and replaced by toluene (2 mL) and the solids were dissolved by heating. Large, yellow, needle-shaped crystals were deposited overnight (0.098 g, 50%).

Preparation of [Zn(meind)₂(tmeda)] (5): 1-Methylindene^[51] (2.21 g, 0.0170 mol) was added dropwise to potassium (0.74 g, 0.0189 mol) in THF (30 mL) and the mixture was stirred at ambient temperature until the potassium had reacted. This solution was then slowly added to ZnCl₂ (1.16 g, 0.0085 mol) in THF (10 mL) and stirred for 12 h. The precipitates were removed by centrifugation and the solution concentrated to 20 mL. TMEDA (0.065 mL, 0.43 mmol) was added to the di(methylindenyl)zinc solution (1.00 g, 0.43 mmol). Crystals of 5 were deposited at ambient temperature overnight (0.078 g, 41%).

Treatment of 5 with NCS: Crystals of 5 (0.10 g, 0.23 mmol) were washed with cold THF and dried in vacuo. By using a glass rod the crystals were ground to a powder and a suspension of NCS (0.45 g) in methanol (10 mL) was added while the crystals were cooled with liquid nitrogen. The mixture was allowed to reach ambient temperature under constant stirring and extracted with hexane (2 × 5 mL). The organic phase was washed with HCl (2.5 M) and the solvent removed in vacuo to afford 1-chloro-1-methylindene (30 mg, 79%). ¹H NMR (400 MHz, CDCl₃): δ = 7.49 (m, 1H, ArCH), 7.24–7.28 (m, 3H, ArCH), 6.66 (d, 1H, H3), 6.39 (d, 1H, H2), 1.87 ppm (s, 3H, CH₃).

Preparation of [Zn(dcp)(succinimidato)(tmeda)] (6): Dimethylzinc (2 M in toluene, 1.0 mL, 2.0 mmol) was added to a solution of succinimide (0.20 g, 2.0 mmol) and 2,6-dichlorophenol (0.33 g, 2.0 mmol) in THF (4 mL) at ambient temperature and the mixture was stirred for 30 min until the evolution of gas had stopped. The solvent was removed in vacuo to yield a white powder. A fraction of the powder (0.05 g) was suspended in diethylether (1.0 mL). TMEDA (0.7 mL) was added the mixture was heated until most of the residue had dissolved. The mother liquor was transferred to a new Schlenk tube and needle-shaped, colorless crystals were deposited overnight (64 mg, 76%). ¹H NMR (400 MHz, CDCl₃): δ = 7.08 (d, 2H, Ar), 6.34 (t, 1H, Ar), 2.70–2.95 (m, 4H, N-CH₂ (succinimidato)), 2.55–2.68 ppm (m, 16H, N-CH₂, N-CH₃ (tmeda)).

Preparation of [Zn(succinimidato)₂(pic)₂] (7): Dimethylzinc (2 M in toluene, 1.0 mL, 2.0 mmol) was added to a solution of succinimide (0.40 g, 2.0 mmol) in THF (4 mL) at ambient temperature and the mixture was stirred for 30 min until the evolution of gas had stopped. The solvent was removed in vacuo to yield a white powder. A fraction of the powder (0.05 g) was dissolved in diethyl ether (1 mL) and 3-methylpyridine (0.50 mL). Colorless crystals were deposited after 2 h at –30 °C (43 mg, 51%). ¹H NMR (400 MHz, CDCl₃): δ = 8.62 (s, 2H, Ar), 8.55 (d, 2H, Ar), 7.62–7.64 (m, 2H, Ar), 7.29–7.32 (m, 2H, Ar), 2.60 (bs, 8H, N-CH₂ (succinimidato)), 2.33–2.36 ppm (m, 6H, CH₃).

Preparation of [Zn(succinimidato)₂(pic)₂]-THF (8): Dimethylzinc (2 M in toluene, 1.0 mL, 2.0 mmol) was added to a solution of succinimide (0.40 g, 2.0 mmol) in THF (4 mL) at ambient temperature and the mixture was stirred for 30 min until the evolution of gas had stopped. The solvent was removed in vacuo to yield a white powder. A fraction of the powder (0.05 g) was dissolved in a mixture of THF (1.5 mL) and 3-methylpyridine (0.2 mL). Block-shaped, colorless crystals were deposited overnight at –80 °C (35 mg, 35%).

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